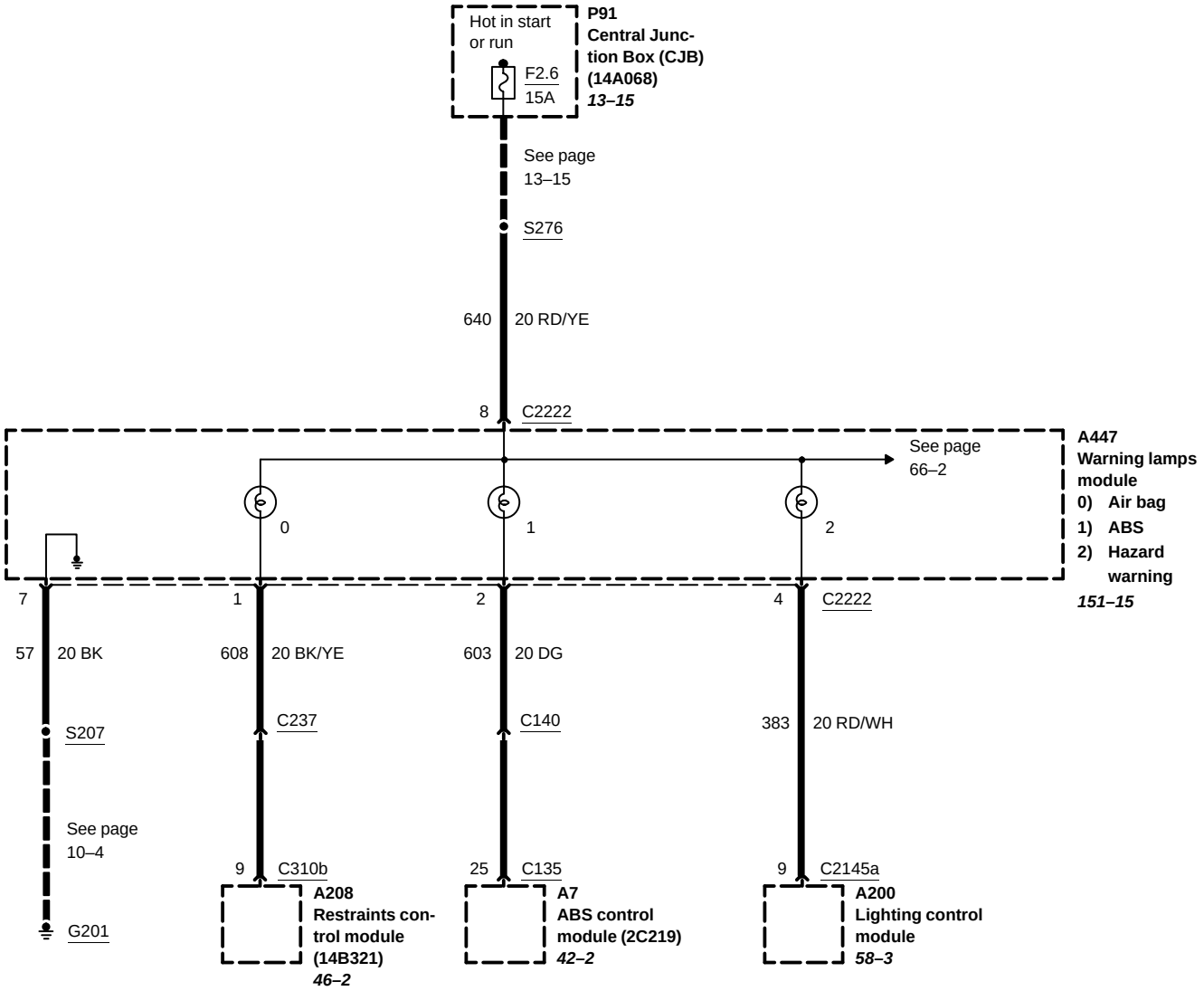electronic cluster